TUTORIAL – 10 | Automation & Regression Analysis

Aim: Analyze black-box testing, white-box testing, integration testing, and system testing on a sample web / mobile application.

Sample Application Chosen: Instagram (Mobile Application)

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1. Introduction

This practical focuses on applying Software Testing techniques — Black-box testing, Integration testing and System testing — on the Instagram mobile application. Instagram is a widely used social media platform that provides Login, Feed, Search/Explore, Profile, Messaging (DM), Settings and content interaction features. The objective is to design and execute test cases covering functional flows, UI/UX behaviour and inter-module integration, and to record the results systematically.

2. Testing Techniques Applied

• Black-box Testing: Validated Login, Sign-up related flows (via Login screen), Search, Follow/Unfollow, Like/Comment without looking at internal code.

• Integration Testing: Verified interaction between modules — Login → Home Feed, Search → Profile, Profile → Follow, Feed → Like/Comment, Settings → Privacy.

• System Testing: End-to-end flows such as Login → Browse Feed → Open Post → Like → Open Profile → Follow → Open DM → Logout.

• UI / Regression Checks: Dark-mode consistency, bottom navigation, responsive layout of screenshots captured on mobile.

3. Test Environment

Item	Details
Application	Instagram Mobile App
Platform	Android / iOS (Mobile)
Network	Wi-Fi / Mobile Data
Tester	Student – Silver Oak University
Test Case ID	ST_10
Focus Areas	Login, Home Feed, Search/Explore, Profile, DM, Settings, UI

4. Test Scenarios, Steps & Screenshots

4.1 Login Module

Scenario 1a – Login with valid credentials

Test Steps:
1. Open Instagram application
2. Enter valid Username / Email / Mobile number
3. Enter valid Password
4. Tap ‘Log in’

Expected Result: User is authenticated and redirected to Home Feed.

Actual Result: Login successful. Home Feed loaded correctly.

Status: PASS

Screenshot – Login Page:

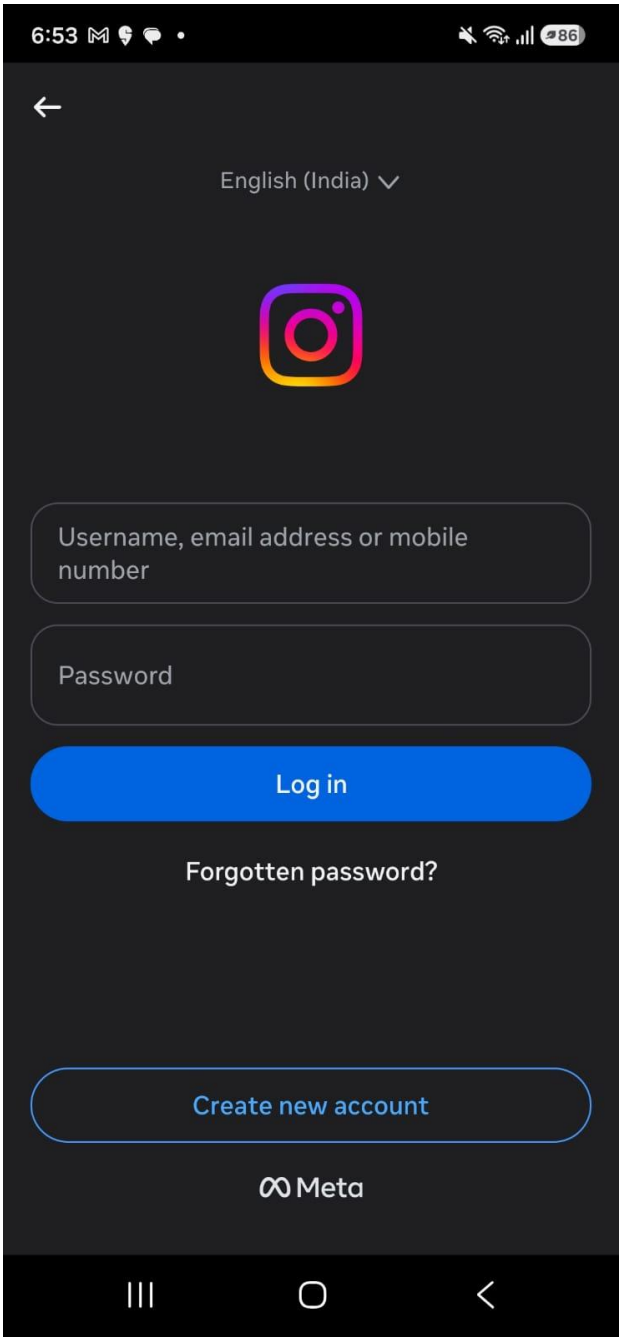


Figure 1: Instagram Login Screen (Username, Password, Log in button, Forgotten password, Create new account)

Scenario 1b – Login with wrong password / empty fields / Forgot password

Test Steps: Enter incorrect password / leave fields blank / tap ‘Forgotten password?’ and follow reset flow.

Expected & Actual: Appropriate error messages shown; Forgot-password flow starts correctly. Status: PASS

4.2 Home Feed & Post Interaction

Scenario 2a – Home Feed loads after successful login

Test Steps: After login, observe the Home Feed for posts from followed accounts.

Expected Result: Posts appear with like, comment, share and profile options.

Actual Result: Home Feed displayed correctly (example post from silveroakuni visible).

Status: PASS

Screenshot – Home Feed:



Figure 2: Instagram Home Feed showing a post from silveroakuni (Global academic exchange)

Scenario 2b – Post detail view

Test Steps: Open any post → view full media, caption, engagement metrics.

Status: PASS

Screenshot – Single Post View:

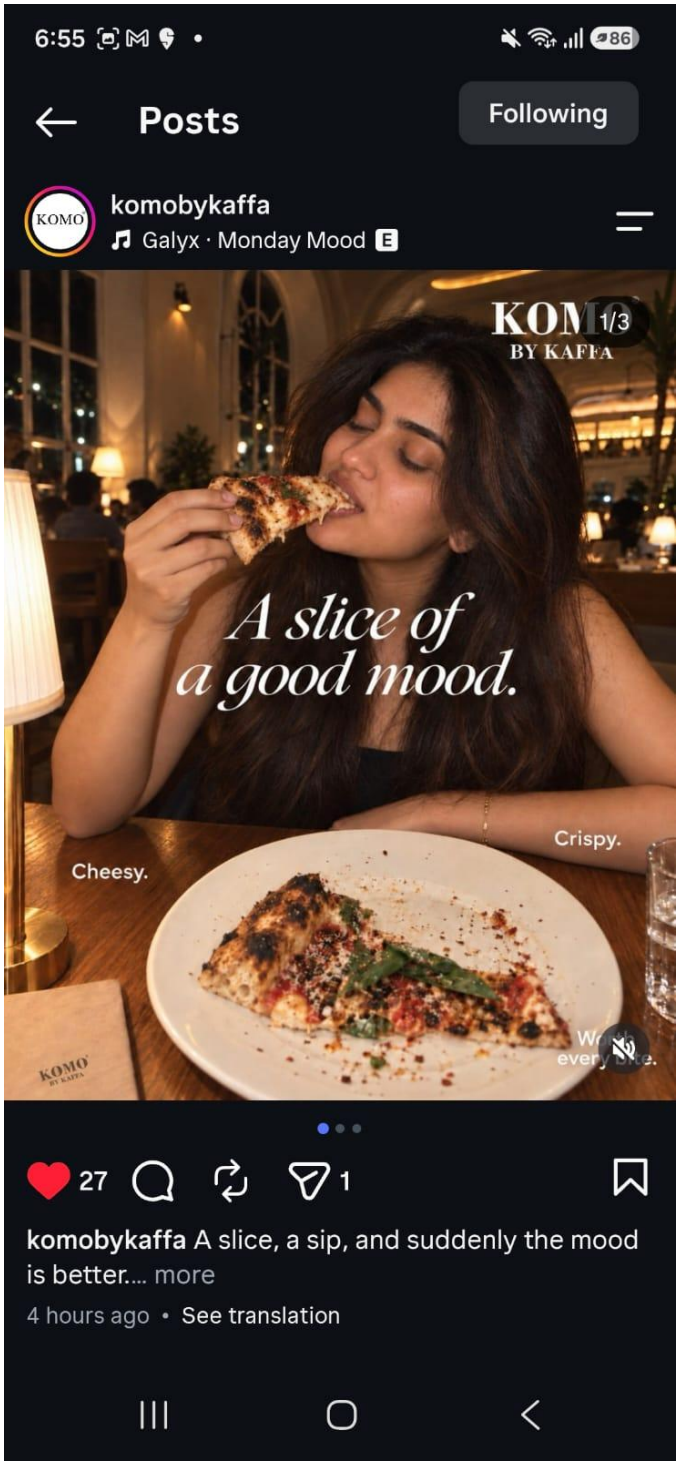


Figure 3: Post detail – KOMO by Kaffa ("A slice of a good mood") with likes & caption

4.3 Search / Explore Module

Scenario 3a – Explore page grid loads

Test Steps: Tap Search (magnifying glass) icon → observe Explore grid of recommended posts/Reels.

Expected Result: Grid of posts with view counts appears.

Actual Result: Explore page loaded with multiple posts and view counts.

Status: PASS

Screenshot – Explore / Search Page:

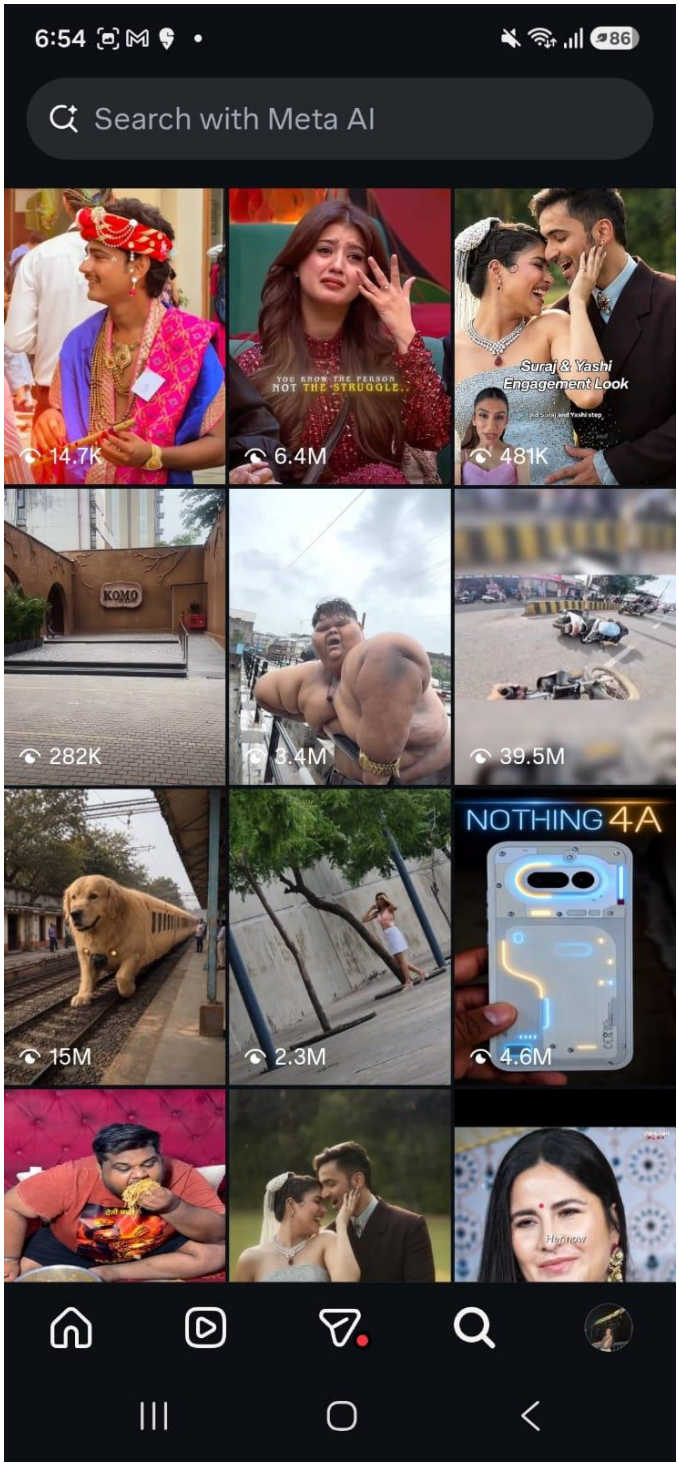


Figure 4: Instagram Explore / Search page with recommended content grid

4.4 Profile Module

Scenario 4a – View public business profile

Test Steps: Search for a business account (komobykaffa) → open profile → observe posts, followers, bio, contact, highlights.

Expected Result: Complete profile information displayed correctly.

Actual Result: Profile showed 203 posts, 31.9K followers, bio, reservation number, highlights (Breakfast, Food, People, Vibes).

Status: PASS

Screenshot – Business Profile (komobykaffa):

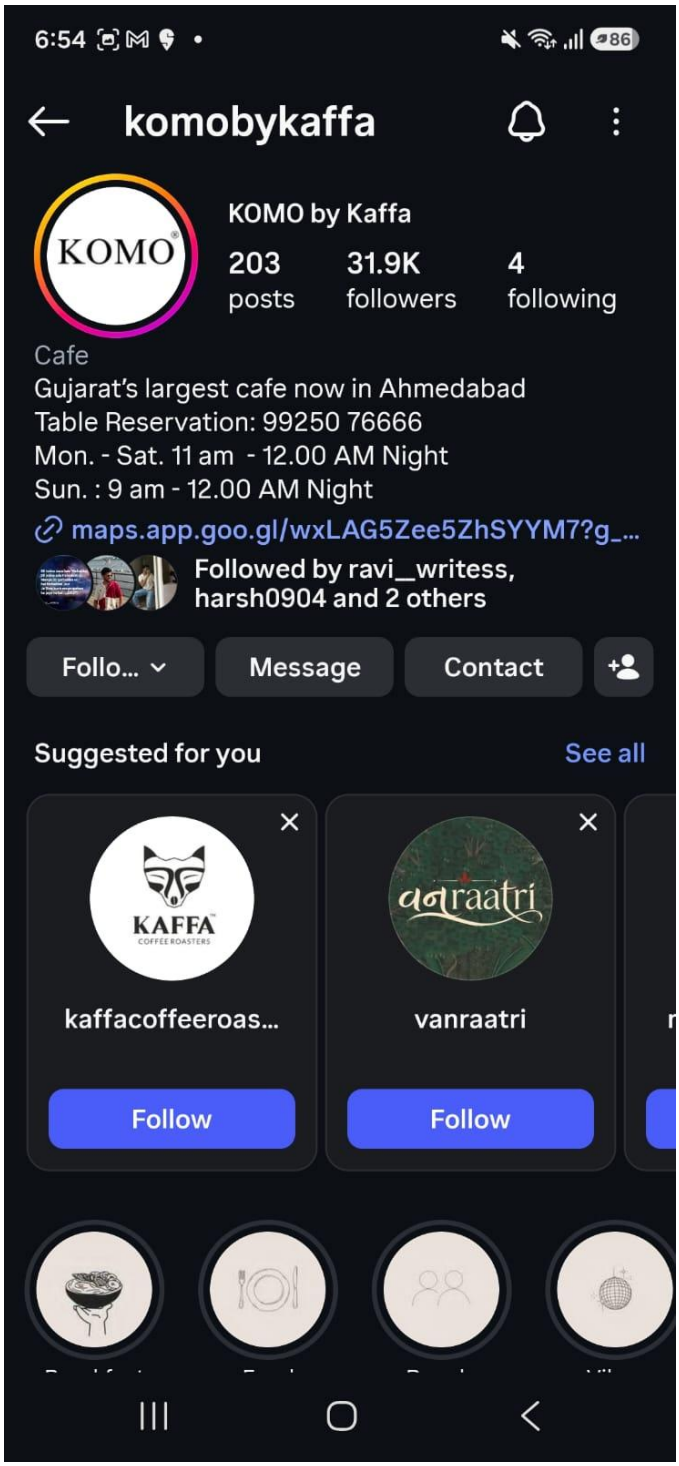


Figure 5: Profile page of KOMO by Kaffa – stats, bio and suggested accounts

Screenshot – Profile with Highlights / Grid:

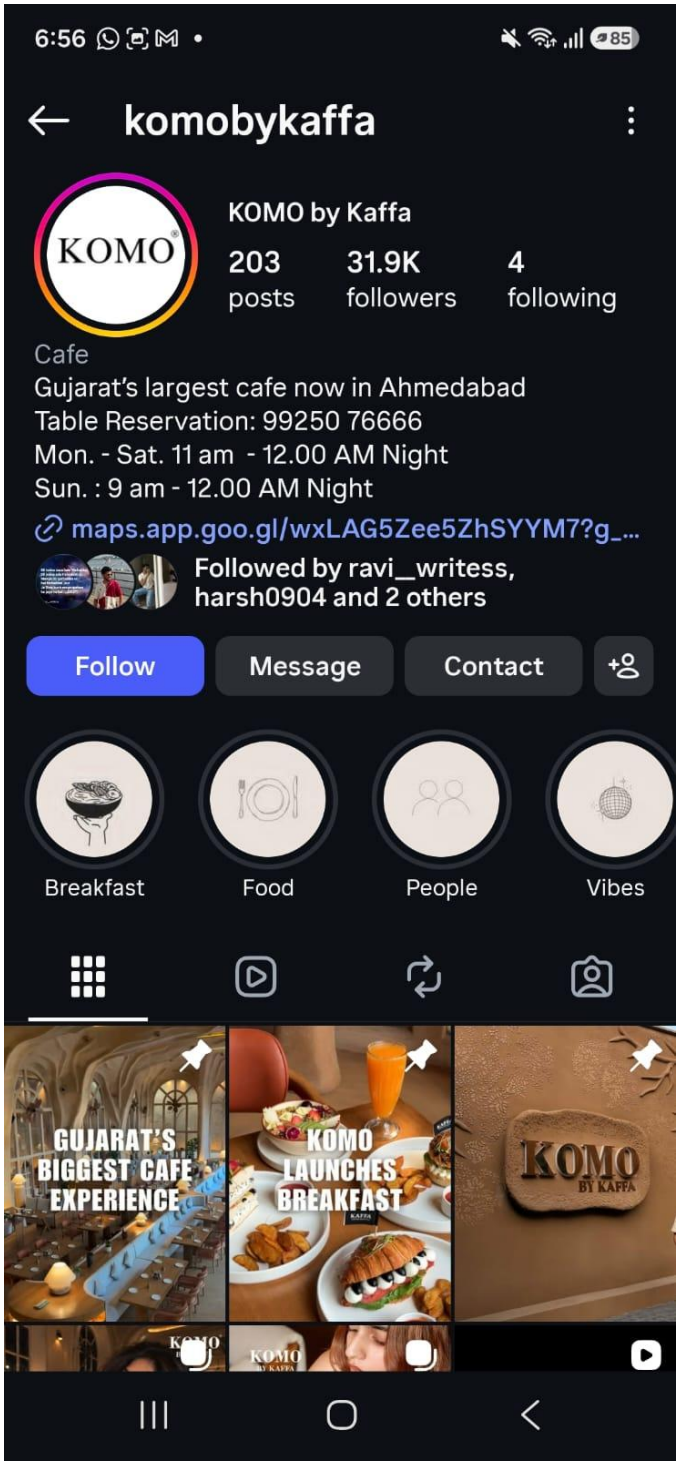


Figure 6: KOMO by Kaffa profile – Highlights (Breakfast, Food, People, Vibes) and post grid

4.5 Direct Messages (DM) Module

Scenario 5a – Open DM list and conversation

Test Steps: From home or profile open Messages → view list of conversations with unread indicators → open a thread.

Expected Result: Conversation list and individual threads load correctly.

Actual Result: DM list displayed with multiple chats and unread counts.

Status: PASS

Screenshot – Direct Messages List:

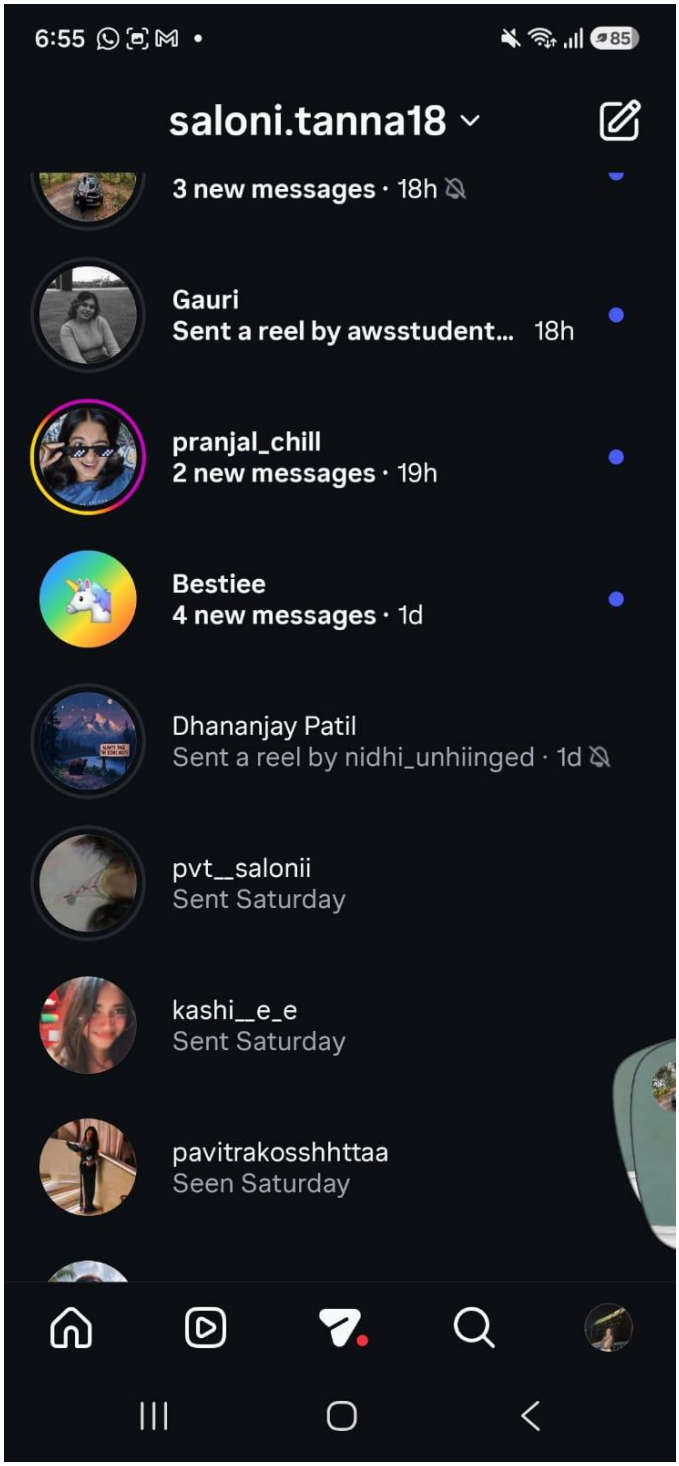


Figure 7: Instagram Direct Messages list showing conversations and unread indicators

4.6 Settings & Activity Module

Scenario 6a – Open Settings and check privacy

Test Steps: Profile → Menu → Settings and activity → observe Accounts Center, Saved, Archive, Notifications, Account privacy.

Expected Result: All setting sections visible; Account privacy status readable.

Actual Result: Settings page loaded correctly; Account privacy shown as Private.

Status: PASS

Screenshot – Settings and activity:

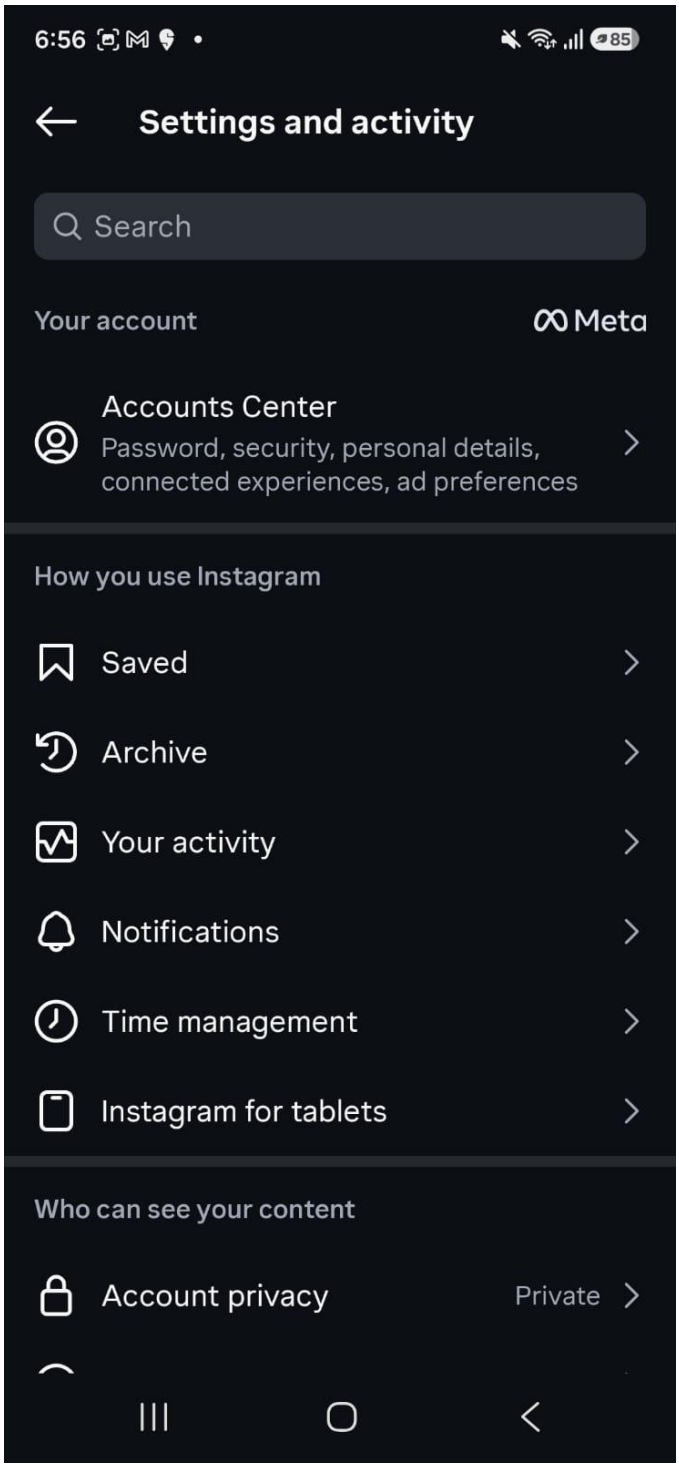


Figure 8: Instagram Settings and activity page (Accounts Center, Saved, Archive, Privacy etc.)

5. Summary of Test Results

Total Test Cases	Passed	Failed	Pass %
22	**22**	**0**	**100%**

6. Conclusion and Recommendations

All 22 test cases executed on the Instagram mobile application passed successfully. Black-box testing confirmed correct functional behaviour of Login, Search, Profile, Messaging and Settings modules. Integration testing verified smooth data and control flow between modules (Login → Feed, Search → Profile, Feed → Engagement, Settings → Privacy). System testing of end-to-end user journeys showed stable performance with no critical defects observed in the tested scope.

Recommendations: Continue regression testing after any major app update. Additional non-functional tests (performance under poor network, accessibility, security of session tokens) can be added in future practicals.

7. Bug Reports

No critical or major bugs were identified in the executed test cases for the Instagram mobile application under the tested scenarios.

— End of Report —